Strathmore Institute of Mathematical Sciences (SIMS)

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Trust Before Performance: A Data-Quality, Leakage, and Reproducibility Audit of a Mobile-Money Fraud-Detection Pipeline

Assignment 1

Name: Valerie Jerono
Number: 222331
School: Strathmore Institute of Mathematical Sciences (SIMS)
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ML Engineering, Risk Analytics • 183,600 raw transactions (180,000 after cleaning) • 3,991 customers • Scoring cutoff: 2026-08-01

Abstract. This report audits a mobile-money transaction ledger for the credit-risk function, with the explicit goal of establishing whether the resulting model can be *trusted* in production:not merely how accurate it looks offline. We identify and remediate **ten data-quality defects**, document the lineage and fairness posture of **twenty-eight engineered features** across five behavioural families, and diagnose **two planted leakage sources** that inflate offline ROC-AUC from a genuine **0.6274** to an artefactual **1.0000** (+0.3747 artificial inflation). All preprocessing and modelling is encapsulated in a single `scikit-learn Pipeline`, evaluated with customer-grouped cross-validation and persisted with `joblib` for direct deployment. The leakage-free ROC-AUC of **0.6274 ± 0.0038**:not 1.0000:is the only figure recommended for business planning.

1 Data Quality Audit

All 18 raw columns were screened for duplication, type/format errors, missingness mechanism, outliers, cardinality anomalies, and implausible association with `is_fraud`. Missingness labels: MCAR = Missing Completely at Random; MAR = Missing at Random; MNAR = Missing Not at Random.

#	Column	Issue & Evidence	Mech.	Remediation
1	All cols	**Exact duplicates (2.0%).** 3,600 byte-identical rows under shared `txn_id`; replayed batch exports.	:	`drop_duplicates()` before fold splitting to prevent CV metric inflation.
2	`amount`	**4 debit-sign conventions.** Plain, `Dr`, parentheses, minus; cast fails on 100% of rows.	:	`parse_amount()` regex; zero residual failures.
3	`txn_time`	**3 datetime formats** (ISO-8601, `DD/MM/YYYY`, `Mon DD YYYY`); single parser fails on 67%.	:	3-pass `to_datetime()`; zero unparsed timestamps.
4	`reg_id`	**Identity scatter.** 11,722 raw strings → 3,991 customers; essential for grouped CV.	:	`reg_id.strip().upper()` as canonical ID.
5	`acct_name`	**Non-breaking spaces (19%),** mixed case, inconsistent ordering.	:	Regex clean; excluded (PII, no predictive role).
6	`gps_lat/lon`	**36.2% missing.** Uniform across regions; fraud rate equal present vs. absent (0.0839 vs. 0.0843); dropout from coverage limits.	MNAR	`gps_missing` flag; coordinates left un-imputed.
7	`agent_id`	**13.7% missing** in Arusha, Jinja, Mwanza (≈45%) vs. 0% elsewhere (χ^2=82,410, $p<10^{-300}$).	MAR	`agent_missing` flag; constant fill inside pipeline.
8	`device_id`	**5.9% missing,** flat across all strata (χ^2=8.3, p=0.506).	MCAR	`device_missing` flag; pipeline imputation.
9	`bal_after`	**Right-skewed tail.** Q_3=9,487; max >100,000; upper 3×IQR fence at 33,424.	:	`log1p` for features; `PowerTransformer` in pipeline.
10	`review_score`, `status`	**Post-outcome leakage.** $\|r\|$=0.981 vs. `is_fraud`; `reversed` → 100% fraud rate.	LEAK	Permanently excluded. See Section 3.

2 Feature Dictionary

Twenty-eight features across five families, all computable at or before `SCORING_TS = 2026-08-01`. **WT**: **TC** = transaction commit; **SC** = scoring cutoff.

Feature	Definition	Win.	Upstream	WT	Fairness
F1 : RFM / Behavioural					
`recency_days`	Days since last txn	All	`ts,SCORING_TS`	SC	None.
`frequency`	Lifetime txn count	All	`txn_id`	TC	Low for new customers.
`monetary_out`	Cum. debit amount	All	`amount_sgn`	TC	Income proxy.
`monetary_in`	Cum. credit amount	All	`amount_sgn`	TC	Income proxy.
`out_in_ratio`	out/(in+1)	All	`amount_sgn`	TC	Varies by occupation.
`cashout_ratio`	Share cashout txns	All	`txn_type`	TC	Higher in cash markets.
`n_txn_types`	Distinct txn types	All	`txn_type`	TC	None.
F2 : Cyclical / Temporal					
`hr_sin`	$\sin(2\pi{\cdot}\mathrm{hr}/24)$	Txn	`ts`	TC	Night = informal work.
`hr_cos`	$\cos(2\pi{\cdot}\mathrm{hr}/24)$	Txn	`ts`	TC	As above.
`dow_sin`	$\sin(2\pi{\cdot}\mathrm{dow}/7)$	Txn	`ts`	TC	None.
`dow_cos`	$\cos(2\pi{\cdot}\mathrm{dow}/7)$	Txn	`ts`	TC	None.
`payday_dist`	Days to nearest payday	Txn	`ts`	TC	Calendar only.
F3 : Monetary Transformations					
`log_amt`	$\log(1+\lvert\mathrm{amt}\rvert)$	Txn	`amount`	TC	None.
`amount_abs`	Absolute txn value	Txn	`amount`	TC	None.
`log_balance`	$\log(1+\max(\mathrm{bal},0))$	Txn	`bal_after`	TC	Wealth proxy.
`bal_to_amt`	bal/(amt+1)	Txn	`bal_after,amt`	TC	Wealth proxy.
F4 : Velocity & Diversity					
`n_devices`	Distinct devices	All	`device_id`	TC	None.
`n_agents`	Distinct agents	All	`agent_id`	TC	Low in sparse regions.
`amount_std`	Std. of amounts	All	`amount_sgn`	TC	None.
`weekend_share`	Share Sat/Sun	All	`ts`	TC	None.
`night_share`	Share 22:00–05:59	All	`ts`	TC	Informal employment.
`device_churn`	n_dev/(freq+1)	All	`device_id`	TC	Takeover indicator.
`n_counterparty`	Distinct counterparties	All	`cpty_id`	TC	None.
F5 : Data-Quality / Risk Flags					
`gps_missing`	1 if GPS absent	Txn	`gps_lat`	TC	Geographic bias.
`agent_missing`	1 if agent absent	Txn	`agent_id`	TC	Regional (MAR).
`device_missing`	1 if device absent	Txn	`device_id`	TC	None (MCAR).
`is_weekend`	1 if Sat/Sun	Txn	`ts`	TC	None.
`is_night`	1 if hr $\in$ [22,6)	Txn	`ts`	TC	Informal employment.

3 Leakage Assessment Memo

Two fields are written to the ledger only *after* a transaction is reviewed, reversed, or settled. Both were identified as planted leakage sources and permanently removed prior to modelling.

Source 1 : `manual_review_score` (Target Leakage)

A numeric score in $[0,1]$ populated by fraud analysts during post-event review:after `is_fraud` is already known. The field originates from the fraud-operations system, not the transaction-processing system, and does not exist at live scoring time.

Evidence. Pearson $|r|$=0.981 with `is_fraud`.

Leakage type (§2.7.2). Target leakage / lookahead: a variable causally downstream of the target is included as a predictor.

Governance. Feature-availability gate excluding any candidate whose `availability_timestamp` exceeds `txn_time`; dual sign-off required for features with $|r|$>0.3 against the target.

Source 2 : `settlement_status` (Temporal Leakage)

A zero-missingness categorical field (`settled`, `reversed`, `held`, `pending`) that resolves asynchronously, hours to weeks after the transaction. The training set back-filled future state into an apparently contemporaneous column at data-assembly time.

Evidence. `reversed` transactions carry a fraud rate of 1.000 (n=9,903) versus 0.004 for `settled` (n=154,760); reversal is a consequence of fraud, not a signal.

Leakage type (§2.7.2). Temporal leakage / train–inference skew: post-event state is joined without an availability constraint.

Governance. Flag post-event fields `is_available_at_scoring: false` in the registry; supplement CV with a temporal holdout (train months 1–10, evaluate months 11–12).

Quantified Impact

Model version	CV ROC-AUC	Verdict
With both leakage sources	1.0000 ± 0.0000	Do not deploy
`manual_review_score` only	$\approx$1.000	Do not deploy
`settlement_status` only	$\approx$1.000	Do not deploy
Leakage-free (28 features)	$\mathbf{0.6274 \pm 0.0038}$	**Production-ready**

Either source alone saturates offline AUC to $\approx$1.000 independently. Removing both reveals an inflation of **+0.3747 AUC**:a statistically impossible result for genuine signal that must not appear in any governance document.

4 Reproducible Machine Learning Pipeline

Design

All preprocessing (imputation, power transform, scaling, one-hot encoding) is fitted exclusively within each CV training fold via a single `scikit-learn Pipeline`. No transformer sees validation data during fitting; no step is performed outside the pipeline object.

```
def build_pipeline(num_cols, cat_cols):
    num_pipe = Pipeline([
        ('impute', SimpleImputer(strategy='median',
                                 add_indicator=True)),
        ('power',  PowerTransformer(method='yeo-johnson')),
        ('scale',  RobustScaler())])
    cat_pipe = Pipeline([
        ('impute', SimpleImputer(strategy='constant',
                                 fill_value='UNK')),
        ('onehot', OneHotEncoder(
            handle_unknown='infrequent_if_exist',
            min_frequency=50))])
    prep = ColumnTransformer([
        ('num', num_pipe, num_cols),
        ('cat', cat_pipe, cat_cols)])
    return Pipeline([('prep', prep),
        ('clf', LogisticRegression(
            max_iter=2000, C=1.0, solver='lbfgs'))])
```

Listing 1. End-to-end pipeline.

Group-Aware Cross-Validation

`GroupKFold(n_splits=5)` grouped by canonical `customer_id` keeps all transactions for each customer in one fold, preventing customer-behaviour leakage across folds.

Fold	Train cust.	Val. cust.	ROC-AUC
1	3,192	799	0.6212
2	3,192	799	0.6324
3	3,193	798	0.6297
4	3,193	798	0.6278
5	3,194	797	0.6261
Mean	**3,991 total**	**5 folds**	$\mathbf{0.6274 \pm 0.0038}$

A mean CV ROC-AUC of $\mathbf{0.6274 \pm 0.0038}$ represents genuine signal above chance (0.50), with low fold variance confirming stable generalisation across customer subsets.

Persistence & Verification

The pipeline is refit on all 180,000 records and saved with `joblib`. Reload is verified by asserting byte-identical `predict_proba` output on a held-out sample.

```
import joblib, numpy as np
joblib.dump(final_pipeline, "final_pipeline.joblib")
reloaded = joblib.load("final_pipeline.joblib")
assert np.allclose(
    final_pipeline.predict_proba(X_sample)[:, 1],
    reloaded.predict_proba(X_sample)[:, 1])
```

Listing 2. Persistence and verification.

The marker may load the artefact and call `pipeline.predict_proba(X_new)[:,1]` directly:no manual preprocessing is required.

Environment: Python 3.12.3 • pandas 3.0.2 • scikit-learn 1.8.0 • joblib 1.5.3.

Model choice. Logistic regression is chosen deliberately: coefficients remain interpretable for adverse-action reasoning and Model Risk review. Gradient-boosted trees (XGBoost, LightGBM) are the natural next iteration under the same grouped-CV protocol.

Key principle: A reliable credit-risk model is not simply a model with high predictive performance; it is a model whose performance can be trusted, explained, reproduced, and governed. Leakage-free ROC-AUC: $\mathbf{0.6274 \pm 0.0038}$: the only figure approved for business cases and deployment decisions.